A CASE-STUDY BASED REPORT

ON

COVID_19 DETECTION FROM CHEST X-RAY IMAGES USING CNN AND VGG16

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by

Name :S. Hasini **roll: 2520030287**

Name :S. Vaibhava Sree **roll: 2520030398**

Under the Esteemed guidance of

Jyothsna Datti

Asst.Professor, Dept. of CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

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ABSTRACT:

COVID-19 is a disease that mainly affects the respiratory system. Chest X-ray images can show changes in the lungs that may be related to COVID-19 infection. Manual examination of a large number of X-ray images can take time and may depend on the experience of the medical professional.

This case study presents a Machine Learning and Deep Learning approach for detecting COVID-19 from chest X-ray images using Convolutional Neural Networks (CNN) and VGG16. The images are prepared and processed before training the models. CNN is used to learn important features directly from the X-ray images, while VGG16 is used as a pre-trained deep learning model through transfer learning.

The main purpose of this project is to study how image-based deep learning can be applied to COVID-19 detection. The performance of the models can be evaluated using measures such as accuracy, precision, recall, F1-score, and confusion matrix. The study also includes Exploratory Data Analysis (EDA) to understand the dataset before model training.

This system is intended as a supporting research tool, not as a replacement for doctors or clinical diagnosis.

INTRODUCTION:

Machine Learning is widely used in different fields, including healthcare. In medical image analysis, Machine Learning can help identify patterns in images that may be difficult or time-consuming to detect manually.

COVID-19 mainly affects the respiratory system. Chest X-ray images may contain visible abnormalities in the lungs of infected patients. Therefore, computer-based analysis of chest X-ray images can be studied as a possible method for assisting COVID-19 screening.

In this project, deep learning techniques are used for image classification. Two approaches are considered:

- CNN – Convolutional Neural Network
- VGG16 – a pre-trained deep CNN model

CNN can automatically learn image features such as edges, shapes, textures, and patterns. VGG16 already contains learned visual features from a large image dataset and can be adapted to the COVID-19 classification task using transfer learning.

The project follows the general steps of collecting the image dataset, preprocessing the images, performing EDA, training the models, evaluating their performance, and comparing the results.

BACKGROUND:

COVID-19 is caused by the SARS-CoV-2 virus. It primarily affects the respiratory system and can cause symptoms ranging from mild illness to serious respiratory problems.

Medical imaging techniques such as chest X-rays are commonly used to examine the lungs. In some COVID-19 cases, abnormalities can appear in chest X-ray images.

Traditional image classification methods require manually designing features. Deep Learning reduces this requirement because the model can learn useful features automatically from images.

Convolutional Neural Network

A CNN is a deep learning model commonly used for image processing. A CNN normally contains:

1. Convolution layers
2. Activation functions
3. Pooling layers
4. Fully connected layers
5. Output layer

The convolution layers extract important features from images. Pooling reduces the size of the feature maps and helps reduce computation. Finally, the fully connected layers perform classification.

VGG16

VGG16 is a popular CNN architecture developed for image classification. It contains multiple convolutional and pooling layers followed by fully connected layers.

Instead of training VGG16 completely from the beginning, transfer learning can be used. In transfer learning, a model that has already learned useful image features is adapted to a new classification problem.

For this project, VGG16 can be adapted to classify chest X-ray images into the required COVID-19 classes.

Overview of the Work Carried Out:

The project is carried out in several stages.

Step 1: Dataset Collection

A chest X-ray image dataset containing the required COVID-19 and non-COVID image classes is used.

Step 2: Data Understanding

The dataset is examined to understand:

- Number of images
- Image classes
- Image dimensions
- Class distribution
- Data quality

Step 3: Image Preprocessing

The images are prepared for the deep learning models.

Common preprocessing operations include:

- Resizing images
- Converting images into a suitable format
- Normalizing pixel values

- Assigning class labels
- Splitting data into training and testing/validation sets

Step 4: EDA

EDA is performed to understand the dataset before training.

The class distribution and sample X-ray images are examined.

Step 5: CNN Model

A CNN model is developed and trained using the prepared X-ray images.

Step 6: VGG16 Model

VGG16 is used through transfer learning. The pre-trained model is adapted for the X-ray classification task.

Step 7: Model Evaluation

The models are evaluated using suitable classification metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Step 8: Comparison

The CNN and VGG16 approaches are compared to understand their performance.

Scope and Responsible-Use Statement:

Scope:

The scope of this project is to study the use of deep learning for classifying chest X-ray images related to COVID-19.

The project focuses on:

- Image classification
- CNN-based learning
- VGG16 transfer learning
- Image preprocessing
- EDA
- Model evaluation
- Comparison of model performance

The project can help students understand how Machine Learning can be applied to real-world healthcare image data.

Responsible Use:

This system should not be considered a medical diagnosis system.

The prediction generated by a Machine Learning model should not be used alone to make medical decisions. A qualified healthcare professional must examine the patient and consider clinical information, laboratory tests, symptoms, and other medical evidence.

The model should be used only for educational, research, and experimental purposes unless it has been properly validated and approved for clinical use.

Problem Statement:

COVID-19 detection from chest X-ray images is a challenging image classification problem. A large number of X-ray images may need to be examined, and identifying useful patterns manually can take time.

The problem is to develop a deep learning-based system that can analyze chest X-ray images and classify them according to the classes available in the dataset, with particular focus on identifying COVID-19-related images.

The study uses CNN and VGG16 to learn visual features from X-ray images and evaluate their ability to perform the classification task.

8. Case-Study Scenario:

Consider a dataset containing chest X-ray images belonging to different classes, including COVID-19-related images and non-COVID images.

Each image is given to the Machine Learning system. The image is first resized and preprocessed. It is then provided to the trained CNN or VGG16 model.

The model analyzes visual patterns in the X-ray and produces a predicted class.

For example:

Input:
X-ray image

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Preprocessing:
Resize → Normalize → Prepare image

↓

Deep Learning Model:
CNN / VGG16

↓

Prediction:
Predicted X-ray class

The prediction can then be compared with the actual class to evaluate the performance of the model.

Central Case-Study Question:

Can CNN and VGG16 effectively classify COVID-19-related chest X-ray images, and which approach provides better performance on the selected dataset?

The study attempts to answer this question by training both approaches and comparing their evaluation results.

Formal Problem Definition

Let the dataset be represented as:

$$\mathbf{D = \{(X_1, Y_1), (X_2, Y_2), ..., (X_n, Y_n)\}}$$

where:

- $\mathbf{X_i}$ represents a chest X-ray image.
- $\mathbf{Y_i}$ represents the corresponding class label.
- **n** represents the total number of images.

The objective is to learn a function:

$$\mathbf{f(X) \rightarrow Y}$$

where the input X is a chest X-ray image and Y is the predicted class.

The CNN and VGG16 models learn patterns from the training images and use these learned patterns to classify previously unseen images.

The final models are evaluated using classification metrics to determine how well they perform on the test data.

Objectives that Guide the Study:

The main objectives of the project are:

1. To study COVID-19 detection using chest X-ray images.
2. To preprocess chest X-ray images for deep learning.
3. To perform EDA on the image dataset.
4. To develop a CNN-based image classification model.
5. To apply VGG16 using transfer learning.
6. To train the models using the prepared dataset.
7. To evaluate the models using suitable performance metrics.
8. To compare CNN and VGG16 performance.
9. To understand the advantages and limitations of deep learning for medical image classification.
10. To develop a system for educational and research purposes.

Assumptions and Limitations:

Assumptions:

The study assumes that:

- The X-ray images are correctly labelled.
- The images are suitable for the classification task.
- The training and testing data represent the classes adequately.
- The preprocessing method is appropriate for the selected models.
- The model can learn useful visual patterns from the available training data.

Limitations:

The project has several limitations:

- The performance depends on the quality and size of the dataset.
- Incorrect or biased labels can affect the model.
- The model may not perform equally well on X-ray images from different hospitals or devices.
- Deep learning models require considerable computational resources.
- High accuracy does not automatically mean that the model is suitable for clinical use.
- X-ray images alone may not provide enough information for medical diagnosis.
- The model should not replace professional medical judgement.

Dataset and Exploratory Data Analysis

About the Dataset

The project uses a chest X-ray image dataset for COVID-19 classification.

The dataset contains X-ray images organized according to their respective classes. The images are used as input to the CNN and VGG16 models.

Before training, the dataset needs to be examined carefully to understand its structure and quality.

Important information to examine includes:

- Total number of images
- Number of images in each class
- Image dimensions
- Image format
- Class distribution
- Presence of missing or corrupted images

Note: The exact dataset size and class counts should be entered from the actual dataset used in your implementation rather than being assumed.

Purpose of EDA

Exploratory Data Analysis is performed before model training.

The main purposes of EDA are:

- To understand the dataset.
- To identify the different classes.
- To check whether classes are balanced.
- To examine sample X-ray images.
- To understand image dimensions.
- To identify possible data-quality problems.
- To determine whether preprocessing is required.

EDA helps ensure that the dataset is properly understood before it is given to the deep learning models.

Preparation Steps Actually Performed:

The following preparation steps are used for the image dataset:

1. Loading the images

The chest X-ray images are loaded from the dataset directories.

2. Assigning labels

Each image is associated with its corresponding class.

3. Resizing

Images can have different dimensions. Therefore, they are resized to a fixed size required by the model.

For VGG16, a commonly used input size is:

224 × 224 pixels

4. Pixel normalization:

Pixel values are scaled or pre-processed so that the model can work with the images more effectively.

5. Training and testing split:

The dataset is divided into training and testing/validation data.

The training data is used to learn patterns, while the testing data is used to evaluate the trained model.

6. Data augmentation:

When appropriate, techniques such as rotation, shifting, zooming, or flipping can be used on training images to increase variation and reduce overfitting.

Only augmentation operations that are medically and technically appropriate for chest X-ray images should be used.

EDA Exploration – Implementation:

EDA can be performed using Python libraries such as:

- NumPy
- Pandas
- Matplotlib
- TensorFlow/Keras
- Scikit-learn

The following activities can be performed:

Class Distribution:

A bar chart can be created to show the number of images in each class.

This helps determine whether the dataset is balanced.

Sample Images:

Sample images from each class can be displayed.

This helps us visually understand the type of X-ray images used in the project.

Image Size

The dimensions of the images can be checked before resizing.

Data Quality

The dataset can be checked for:

- Missing files
- Corrupted images
- Incorrect labels
- Duplicate images
- Unusual image formats

Example EDA Process:

Dataset

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Load image paths

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Identify classes

↓

Count images per class

↓

Display sample images

↓

Check image dimensions

↓

Check data quality

↓

Preprocess images

Summary of EDA Findings:

The EDA provides an understanding of the image dataset before model training.

The major observations to report from the actual dataset are:

- The dataset contains chest X-ray images belonging to multiple classes.
- The images may have different original dimensions and therefore require resizing.
- The number of images in each class should be checked to identify class imbalance.
- Sample images help verify whether the images are readable and correctly labelled.
- Preprocessing is required before providing the images to CNN and VGG16.
- Data augmentation may be useful when the training dataset is limited.

For the final report, replace the general observations above with your actual image counts, class distribution, and EDA graphs.

Algorithm Selection and Implementation:

CNN:

A Convolutional Neural Network (CNN) is selected because it is designed to work effectively with image data.

A basic CNN contains:

Input Image

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Convolution Layer

↓

Activation Function

↓

Pooling Layer

↓

Convolution Layer

↓

Pooling Layer

↓

Flatten

↓

Fully Connected Layer

↓

Output Layer

Working of CNN:

The convolution layers detect visual features such as edges, shapes, textures, and patterns.

Pooling layers reduce the spatial size of the feature maps.

The extracted features are then passed to fully connected layers.

The final layer produces the predicted class.

VGG16:

VGG16 is a deep CNN architecture that contains multiple convolutional layers and pooling layers.

Instead of training all layers from the beginning, transfer learning can be used.

The process is:

Pre-trained VGG16

↓

Remove/replace original classification layer

↓

Add new classification layers

↓

Train using chest X-ray images

↓

COVID-19 classification:

VGG16 is useful because the pre-trained network has already learned general visual features. These features can be adapted to the chest X-ray classification problem.

CNN Implementation Process

The CNN implementation follows these steps:

1. Load the chest X-ray dataset.
2. Assign labels to the images.
3. Resize the images.
4. Normalize the image values.
5. Divide the dataset into training and testing/validation sets.
6. Create the CNN architecture.
7. Compile the model.
8. Train the model.
9. Validate the model.
10. Test the model.
11. Calculate evaluation metrics.
12. Analyse the results.

VGG16 Implementation Process:

The VGG16 implementation follows these steps:

1. Load the pre-trained VGG16 model.
2. Use the required input image size.
3. Remove or replace the original classification layer.
4. Add new classification layers suitable for the dataset.
5. Freeze selected pre-trained layers initially.
6. Train the added layers using the chest X-ray dataset.
7. Validate the model.
8. Evaluate the model using test data.
9. Calculate performance metrics.
10. Compare the result with the CNN model.

Model Evaluation:

The models can be evaluated using the following measures.

Accuracy:

Accuracy represents the percentage of correctly classified images.

Accuracy = Correct Predictions / Total Predictions

Precision:

Precision shows how many images predicted as a particular class are actually from that class.

Recall:

Recall shows how many actual images of a particular class were correctly identified.

For medical screening problems, recall can be especially important because missing a positive case can be serious.

F1-Score:

F1-score combines precision and recall into a single measure.

Confusion Matrix:

A confusion matrix shows:

- True Positive
- True Negative
- False Positive
- False Negative

It helps understand the types of mistakes made by the model.

Comparison of CNN and VGG16:

Metric	CNN	VGG16
Accuracy	74.29%	74.66%
Precision	76%	77%
Recall	74%	75%
F1-Score	73%	73%

Conclusion:

This case study demonstrates the application of Machine Learning and Deep Learning to chest X-ray image classification for COVID-19-related detection.

CNN is useful because it can automatically learn important visual features from images. VGG16 provides another approach through transfer learning, where a pre-trained deep learning model is adapted to the X-ray classification problem.

The project includes important Machine Learning steps such as dataset preparation, EDA, image preprocessing, model training, testing, and evaluation. Metrics such as accuracy, precision, recall, F1-score, and confusion matrix can be used to understand model performance.

The comparison between CNN and VGG16 helps us understand the advantages of different deep learning approaches for image classification.

Overall, the project demonstrates how deep learning can be applied to healthcare image analysis. However, the developed system should be treated as an educational and research-based tool and not as an independent medical diagnosis system. Professional medical judgement and appropriate clinical testing are required for actual diagnosis.